Bin Hu

Systems Software & Networking | Ph.D. Student, USC

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EDUCATION

University of Southern California (USC)

Ph.D. in Electrical and Computer Engineering | Aug. 2023 - 05/2028 (Expected)

Prior doctoral studies: University of Minnesota, CSE, Aug. 2022-Aug. 2023; transferred with advisor.

Southeast University (SEU)

M.Sc. in Computer Science and Engineering | Sept. 2019 - Jun. 2022

B.Eng. in Computer Science and Engineering | Sept. 2015 - Jun. 2019

TECHNICAL SKILLS

Programming: Python, C/C++, Java, MATLAB.

Systems & networking: Linux, WebRTC, Android, Raspberry Pi, TCP/UDP sockets, multipath networking.

Development & analysis: Git, Docker, Wireshark, tcpdump, PyTorch, GNURadio.

ENGINEERING & RESEARCH EXPERIENCE

University of Southern California / University of Minnesota

Ph.D. Student | Aug. 2022 - Present

- **NIER / Android neural video:** Contributed to rebuilding NIER's neural video-conferencing model for Android phones.
- **WebRTC video measurement:** Developed a WebRTC-based system to record real-time video-call quality over satellite networks; used Docker in the project.
- **Multipath networking / Raspberry Pi:** Built client, server, and proxy components for a Raspberry Pi multipath system; increased throughput by 50-80% and reduced packet-loss rate to about one-tenth of a single satellite connection.
- **Android network measurement:** Contributed to Android client development for network experiments measuring latency and packet loss; used Python, C++, and Java across these systems projects.
- **Data pipelines & PyTorch prediction:** Collected and synchronized network measurements and satellite information, preprocessed training data, and built PyTorch models for network and satellite prediction.
- **Self-supervised network modeling:** Contribute to Net-JEPA, a 4M-parameter world model for adaptive streaming and congestion control; framework evaluation uses 450,000+ Puffer streams and 20-trace transfer calibration.

SUTD | Research Assistant | May 2021-Apr. 2022; Jul. 2021

- Developed interpretable ML models for modulation classification and implemented an AI-based software modulator.

Southeast University | Graduate Researcher | Sept. 2019-Oct. 2020; Jun. 2021

- Designed and evaluated a multidimensional LoRa demodulator and implemented a fine-grained self-correlation collision detector (ICNP 2020; WASA 2021).

SELECTED PUBLICATIONS

Predicting Connectivity and Link Performance in Dynamic LEO Satellite Networks. **ACM HotMobile, 2026.** Bin Hu et al.

NN-Defined Modulator: Reconfigurable and Portable Software Modulator on IoT Gateways. **USENIX NSDI, 2024.** Jiazhao Wang et al. (co-author: Bin Hu)

INDUSTRY EXPERIENCE

Guangzhou Desheng Artificial Intelligence Technology Co. Ltd.

Algorithm Engineer (Intern) | Aug. 2018

- Developed camera-based assisted driving systems
- Implemented driver behavior detection and warning algorithms